

Foundations of Data and AI Literacy

Participant Guide

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Class activities



Exercise: Data Interaction

Goal: To become more aware of the data or information you encounter daily and how it influences your decisions.

1. Consider the data you interact with regularly

- Take 5 minutes to think about the information you use regularly. This could include anything from news articles and social media posts to weather reports, maps, etc.
- Write down a list of 3-5 specific types of information/data that you rely on. List your responses in the "Data you interact with regularly" column

2. Identify characteristics of that data

- For each item on your list, analyze its characteristics by considering the following questions:
    - What type of data is it?
    - Is it structured, semi-structured or unstructured?
    - What is the source of this information?
    - Is it big data or small data?
- List your responses in the "Characteristics" column.

3. Reflect on your relationship with data

- How do you use this data to make decisions or support your choices?
- What data sources do you trust, and why do you consider them reliable?
- List your responses in the "Relationship with data" column.

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Data you interact with regularly	Characteristics	Relationship with data
Sell out data	Quantitative discrete type, Structured Comes from retailers and is small data. It has a semi constant flow, but the data is little enough to work with in simpler tools.	Use it to get clear view of trends in the data. They are reliable as they are structured and constant.
Pricelevels of products at different retailers over time	Quantitative continuous data, semi-structured. Comes from scraping the internet and then we buy it (external data). This is big data as it is constant stream at thousands of websites and it can be in any format (text, number value, image)	This used to support the sell-out data and to see general trends in pricing. It is semi-reliable as it is semi structured and big data. In general there is no trouble with the data, but at the data should be sanity checked.
App downloads	Quantitative discrete type, Semi-Structured Comes from scraping app store and google play and is small data. It has a constant flow, but the data is not too big and is a clear structure.	This is to see market share. It is reliable as it is an external source and a more structured semi-structure.
Campaign information	Qualitative nominal data, semi-structured. Is gathered from the retailer, it is small data as they manually pick what to send and we manually look at it.	To analyze the campaigns. It is partly reliable as it is a more structured semi-structure and small data, but the source is not as viable
Reactions to campaigns and products	Qualitative ordinal data, unstructured. Is gathered from the web, it is big data as it constantly changes (flows) in big quantities in many places.	To analyze the campaigns. It is not reliable as it is big data that not yet is handled very well, this makes it hard to know if all data has been gathered and processed.



Exercise: Data analytics case study

Goal: To examine the scenario and classify the provided questions according to the four stages of analytics: descriptive, diagnostic, predictive, and prescriptive.

Scenario:

A prominent chain of fitness centers called Titan Fitness is experiencing an unfavorable trend due to its declining membership renewals and a rise in customer churn. Despite the chain's strong brand reputation, cutting-edge equipment, and wide range of classes, Titan Fitness is struggling to retain customers in an increasingly competitive market. Recognizing the urgency of this situation, the marketing team has decided to investigate the issue and develop a data-driven strategy to revitalize the business and restore customer loyalty.

The task force has access to a wealth of data from Titan Fitness's various systems:

- Customer membership data: Detailed records of all members, including demographics, membership type (monthly, annual, etc.), sign-up date, renewal history, and personal training sessions.
- Usage data: Records of gym visits, class attendance, and equipment usage collected through membership card swipes and booking systems.
- Customer feedback data: Surveys, online reviews, and customer service interactions capture member satisfaction, complaints, and suggestions.
- Financial data: Revenue and expenses related to membership fees, personal training, merchandise sales, and operational costs.
- Marketing data: Data on marketing campaigns across various channels (social media, email, local partnerships), including costs, reach, and conversion rates.

- Competitor data: Information on competing gyms and fitness studios, including their membership pricing, class offerings, and online reviews.
- Market trends data: Reports and analyses on fitness industry trends, emerging technologies, and changing customer preferences.

The following questions were addressed through the application of data analytics. For each question, identify the data analytics stage where you would expect to find a solution or gain the necessary insights.

Questions	**Data analytics stage**
1. Why are membership renewal rates declining?	Diagnostic
2. What strategies can be implemented to improve member retention?	Presciptive
3. Which members are most likely to cancel their memberships in the next quarter?	Predictive
4. What is the average membership length before cancellation?	Descriptive
5. Are certain locations experiencing higher churn rates than others? If so, why?	Descriptive and then Diagnostic
6. Which membership types (monthly, annual, etc.) have the highest and lowest renewal rates?	Descriptive
7. What are the most common reasons cited for membership cancellations in customer feedback?	Descriptive
8. What will the projected membership churn rate be in the next year?	Predictive

9. Are there specific demographics or membership types with significantly higher churn rates?	Descrpitive
10. Is there a relationship between marketing campaign spend and membership sign-ups?	Diagnostic
11. How will changes in membership fees likely affect renewal rates?	Predictive can be prescriptive if you want know what will happen
12. What is the overall customer satisfaction rating based on surveys and online reviews?	Descriptive
13. What are the most popular times of day and days of the week for gym usage?	Descriptive
14. What changes should be made to our marketing campaigns to better target potential and existing members?	Prescriptive
15. Should we expand into new markets or demographics to reach a wider audience?	Prescriptive



Exercise: AI validation

Goal: Experience AI "hallucinations" firsthand without using proprietary or external documents.

Open any approved text LLM (i.e. Copilot, Claude, or Gemini).

Type the prompt: "**List 5 landmark legal cases or academic studies regarding AI data privacy in the workplace, including the year and a two-sentence summary for each.**"

Choose one cited case/study and quickly verify its existence via a standard browser search.

Next, reflect on your experience and jot down your thoughts based on the following questions:

- Did the model generate real sources, or did it invent plausible-sounding titles/authors?

  It used real sources but not all were academic papers/studies, some were news articles/blogs (it didn't seem that the new/blogs were based of academic research. the date was correct in al cases and names as well.

- What workplace risks occur if employees accept AI-generated references without double-checking?

  Even if the AI uses factual information/ find a source, it doesn't mean that the source is accurate. We have been teached to scope the objectivity and credibility, but this type of source criticism is not really part of the system.



Exercise: AI and ML reflections

Goal: To reflect on AI and its components and examine potential applications and implications in your work/field.

Take 10 minutes to reflect on the applications and implications of AI and ML methods for a data project and log your thoughts based on the questions below. When logging your thoughts in the table, think about your own workplace or industry.

Question	Response / Thoughts / Notes
What AI concept did you find most interesting today, and why?	Text mining and graph analysis was new to me. I had partly stumbled upon text mining before, but did not know it was semi-supervised. It would have been nice a bit more technical aspects, but that is not this course.
What task or process could be improved with AI? How will it improve things? (Think about repetitive tasks in your day-to-day work you could automate.)	I liked the idea of mage generation to display thought. I can sometimes have feeling that drawing things out could explain the issue, but can't get it into words and I am to bad at drawing. My problem here is that I do not want to share most information as it is classified to diffeent degreess.

Think about your work or industry. Can you describe a scenario where you would need to use techniques like clustering, classification, regression, text mining to solve a problem or gain insights? Provide a use case for at least one technique and explain why that technique is suitable for this situation.	Text mining: review websites to see peoples responses to products and campaigns. Clustering: customer segmentation, especially what different campaigns and retailers should focus on. Classification: What material has been posted during campaigns, did it include everything it should (product picture, lifestyle image, brand name, campaign deal). Regression: what is the forecased value for this campaign.
Can you think of some tasks for which generative AI could be used?	Maybe product descriptions or marketing material. I do however believe that consumers most often can recognize when work is generative AI's. And I do not think this will help or support neither sales nor image.



Exercise: Detect errors

Goal: To critically examine a sample dataset and identify any gaps or inconsistencies that might affect its reliability or usefulness.

Take 5 minutes to evaluate the dataset and note down the inaccuracies in it.

	A	B	C	D	E	F
1	**Name**	**Last Name**	**DOB**	**Country**	**Country**	**Status**
2	Andrew	Jackson	11/4/87	Argentina	Buenos Aires	Citizen
3	Natalie	Walker	2/12/82	France		
4	Sam	Williams	3/30/80	Argentina	Mendoza	Permanent resident
5	Lilian	Perez	2-Jan-99	France	Nantes	Permanent resident
6	Sarah	Sanchez	1/8/78	Germany	Munich	Citizen
7	J	McAdam	7/18/00	England	London	Permanent resident
8	Oliver	Oliver	12/21/90	Ireland		Citizen
9	Natalie	Walker	2/12/82	France	Paris	
10	Iris	Nelson	10/20/95	India	Mumbai	Citizen
11	Natalie	Roberts	9/30/90	Germany	Munci	Citizen
12	Nina	Rose Turner	1/19/01	UK	London	Citizen
13	Tara	Morris	2/20/63	Delhi	India	Citizen
14	Paul		8/24/03	America	Buenos Aires	Permanent resident
15	PAULINE	BROOKS	7/7/98	India	Banglore	Permanent resident
16	Natalie	Walker	2/12/82	France	Paris	Citizen

Record your responses in the space below:



Exercise: Data management reflections

Goal: To reflect on your organization's data management practices across three core components: data collection, storage and access.

Take 10 minutes to reflect on each component in the context of your organization's data management infrastructure. Use the questions in the table below below as thought prompts and document your responses:

Question	**Response / Thoughts / Notes**
Where do you get most of your data from? What methods are used to collect data (automated tools, manual entry, APIs, web scraping, etc.)?	
What are your primary sources of data?	

Who is responsible for data quality and security?	
Where do you keep all your data? What type of storage systems are used (databases, data warehouses, data lakes, cloud storage)?	
Is it easy to find the information you need? Why or why not?	

If you need information that is stored in a different department, what do you do?	
Are there any security measures in place that you are aware of (encryption, access controls, authentication)?	



Exercise: Data tasks

Goal: To explore a variety of data-related tasks performed by others and identify those that might be relevant to your work.

Review the following list of data tasks and select the ones that you have performed during your professional career.

Note: If none of the tasks resonate with your experience, that's perfectly fine!

- Wrangling the data
- Managing the data
- Creating basic analyses and visualizations
- Asking the right questions from the data
- Building complex predictive models
- Interpreting the results critically and communicating them well
- Ensuring data integrity across different data sources
- Building out additional data warehouses as needed
- Maintaining data pipelines and access
- Requirements engineering
- System design
- Implementation and testing new AI technologies
- Maintenance, support, troubleshooting AI systems
- Planning out people and resources for projects
- Communicating data insights to executives and stakeholders
- Prompt engineering and contextual querying
- Validating AI-generated outputs for hallucinations
- Applying data privacy safeguards when using AI tools



Exercise: Data tasks and tools reflections

Goal: To encourage critical thinking about the various roles, skills, and technologies contributing to your organization.

Take 10 minutes to carefully consider the different types of personnel and tools that your organization offers. Use the questions in the table below below as thought prompts and document your responses:

Question	**Response / Thoughts / Notes**
Who do you usually work with on projects or tasks?	
How do you communicate and collaborate with others in your department? Is there a collaboration tool that you use?	

How effectively do different teams in your organization collaborate on data projects?	
What tools/technologies do you use most often in your work?	
Do you find these tools easy to use? Why or why not?	



Exercise: Data project outline

Goal: To analyze a business scenario and prepare a project outline.

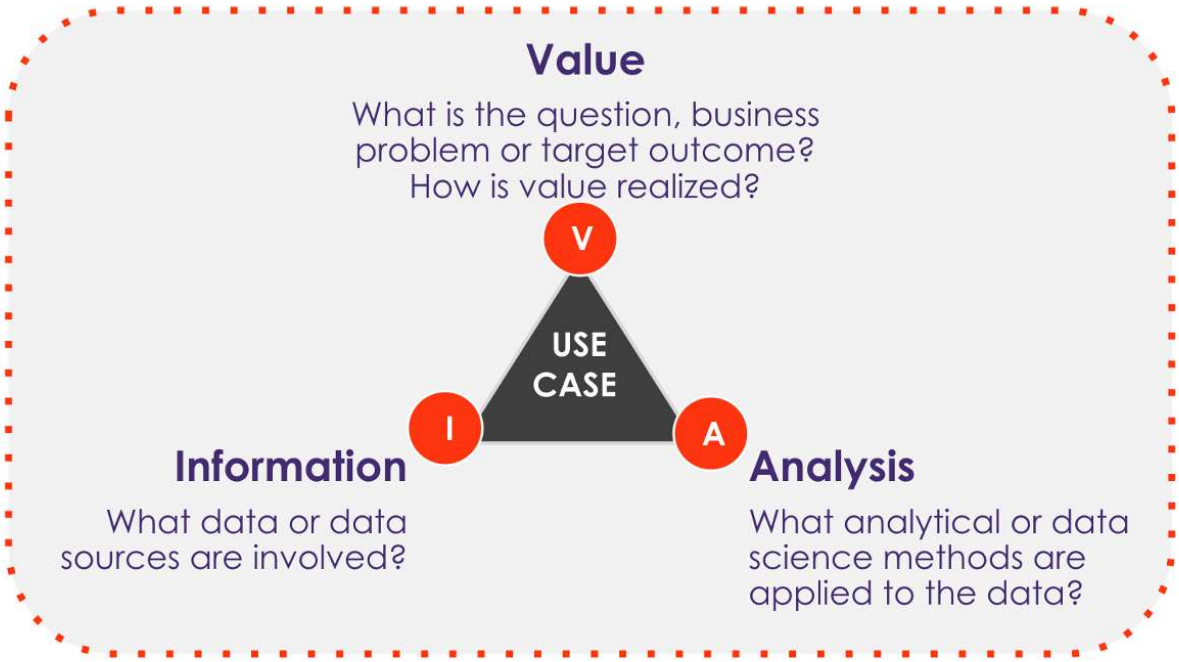
Analyze the scenario below and brainstorm a solution that considers the Value, Information, and Analytics components.

Record your responses in the table below.

Scenario:

A project team working on a high-priority initiative is struggling to meet expectations. Deadlines are consistently missed, and team morale is at an all-time low. Team members feel overwhelmed by the workload, and miscommunication is exacerbating the situation. Their manager recognizes the need for immediate intervention to restore productivity and cohesion. By leveraging data, they aim to identify root causes and implement effective solutions.

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- **Value:** Discuss the scenario and ensure you understand the problem statement and clarify any ambiguities.
- **Information:** Brainstorm and list the key stakeholders, data requirements, and potential key questions related to the problem.
- **Analysis:** Analyze the scenario using the identified information and questions. Think about different perspectives, explore possible solutions, and evaluate their potential impact.

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	How would you simplify or break this down?	What questions do you have?
Value (Understanding and defining the problem)		
Information (Categorize the different types of data you have)		
Analysis (What data analytics methods, if any, with the identified stakeholders would you apply to gain a deeper understanding of the situation?)		

Additional blank space to jot down notes/comments, if needed.

Additional resources

Data and AI glossary

Agentic Workflows / AI Agents - AI systems capable of planning, using tools (like searching web/databases), and executing multi-step workflows autonomously under human supervision.

Algorithm – An algorithm is a series of steps to accomplish a task (a set of directions that gets you from point A to point B, if you will). It can be a thought exercise, a series of mathematical expressions, or a piece of code (or pseudo-code). Algorithms are used in everyday life, and many industries. They are an essential building block of data science, analytics, and any other quantitative field.

API – An application programming interface (API) is an "entryway" to a computer system (such as a database) that allows you to access, retrieve, and edit its components. Most often APIs are used as data-communication mediums between applications. They are an essential part of scalable data-centric applications, research projects that are built around data, and anything else that requires automated data access.

Artificial Intelligence – Artificial intelligence (AI) is the apparent ability of machines to act "intelligently."

Big Data – Big Data is a term that describes a large volume of data—both structured and unstructured. It is produced in such quantities that it cannot be ingested or processed by a single machine at once (even a large machine like a supercomputer), so special tools and techniques are needed to process and store it.

Note: This term is often misused; most data that is called Big Data is not really that big. True Big Data is produced by things like Google Searches, satellite imagery of the Earth, climate simulations used by weather services, sensor data generated by cell towers, etc.

Classification – Classification is a supervised machine learning method where the output variable is a category, such as "yes" or "no."

Clustering – Clustering is an unsupervised machine learning method used to discover groupings that are inherent in the data.

Clickstream – A clickstream is a record of a user's activity on the Internet, including every Web site that the user visits, how long the user was on a page or site, in what order the pages were visited, and newsgroups that the user participates in and even the e-mail addresses of mail that the user sends and receives.

Data Governance – The management of the overall quality, integrity, relevance, and security of available data.

Data Lake – A data lake is a system or repository of data stored in its natural/raw format, usually object blobs or files.

Note: Data lakes and data warehouses are both widely used for storing big data, but they are not interchangeable terms. A data lake is a vast pool of raw data, the purpose for which is not yet defined. A data warehouse is a repository for structured, filtered data that has already been processed for a specific purpose. (Source: https://www.talend.com/resources/data-lake-vs-data-warehouse/)

Data Mining – Data mining is a study of extracting information from structured/unstructured data taken from various sources.

Data Visualization – Any attempt to make data more easily digestible by rendering it in a visual context (e.g., charting, graphing, etc.).

Data Warehouse – A data warehouse is electronic storage of a large amount of data, which is designed for query and analysis. It is typically used to connect and analyze data from heterogeneous sources.

Note: Data lakes and data warehouses are both widely used for storing big data, but they are not interchangeable terms. A data lake is a vast pool of raw data, the purpose for which is not yet defined. A data warehouse is a repository for structured, filtered data that has already been processed for a specific purpose. (Source: https://www.talend.com/resources/data-lake-vs-data-warehouse/)

Data Wrangling – Data wrangling is the process of cleaning, structuring, and transforming raw data into a usable format for analysis, essentially taking messy, inconsistent data and making it organized and ready for further use, often involving tasks like handling missing values, removing duplicates, and standardizing data formats; it's also known as data munging.

Database – A database is a structured collection of data. The collected information is organized in a way such that it is easily accessible by the computer. Databases are built and managed by using database programming languages.

Dataset – A dataset is a collection of data, which is organized into some type of data structure. Several characteristics define a dataset's structure and properties. These include the number and types of the attributes or variables, and various statistical measures applicable to them, such as standard deviation.

Deep Learning – Deep learning is a branch of machine learning that uses deep artificial neural networks. Artificial neural networks are types of machine learning algorithms that are built based on the idea of how a brain works with its neurons connected to each other and transmitting signals. Artificial neural networks are usually built in layers, those that have 3+ layers are considered "deep" and fall into the deep learning bucket.

Generative AI - AI models designed to generate new content (text, images, code, audio) by learning patterns from massive training data.

Graph Analysis – Also known as **network analysis**. Graph analysis is an analysis of structures that model pairwise relationships between objects. The objects in graph analysis are called nodes. Their relationships (a.k.a. connections) are called edges. Graph analysis can be used across many industries, but the most common use case is analysis of social networks.

Hallucination - A phenomenon where a generative AI model outputs confident but factually incorrect or fabricated information.

Hypotheses – An educated guess or statement that a data scientist makes about a potential relationship or pattern within a dataset.

Large Language Model (LLM) - Deep learning models built on Transformer architectures trained on massive text datasets to understand and generate human language.

Machine Learning – Machine learning is the computational process wherein a machine "learns" and adjusts its behaviors based on feedback from data. Usually manifesting as an adaptable algorithm, machine learning helps computers predict outcomes without explicit human input.

Model – A model is a simplified replica of any real-life phenomenon / object at smaller scale. In quantitative disciplines, a model is usually a mathematical description of such a phenomenon / object.

Example: An example is a model to predict the level of education based on age. In a simplified way it looks like this:

Level of education = some quantity + some quantity * age (a linear model)
Although in real life there are many more factors and variables, this model could potentially display a general trend.

Open Data – Open data is data that can be freely used, shared and built-on by anyone, anywhere, for any purpose.

Regression – Regression is a supervised learning method where the output variable is a real value, such as "amount" or "weight" and the input variable(s) is a real value, such

as “size” or “height.” The output variable depends on the input variable(s). The relationship between the input and the output are usually recorded in a mathematical model.

Retrieval-Augmented Generation (RAG) - A technique that connects LLMs to an organization's secure internal database/documents to generate grounded responses without retraining the underlying model.

Structured Data – Structured data is data that has been organized into a formatted repository, typically a database, so that its elements can be made addressable for more effective processing and analysis.

Supervised Learning – Supervised learning is a type of learning in which we teach or train the machine using data that is well labeled, meaning some data is already tagged with the correct answer class (or group to which it belongs).

Training data – A collection of data examples used to train machine learning (ML) models

Text Mining – Text mining is the process of converting unstructured text data into meaningful and actionable information.

Unstructured Data – Unstructured data is information that either does not have a pre-defined data model or is not organized in a pre-defined manner.

Unsupervised Learning – Unsupervised learning is the training of machines using information that is neither classified nor labeled and allowing the algorithm to act on that information without guidance.

Vector Database - A database optimized for storing numerical coordinates (embeddings) that represent the semantic meaning of unstructured data.

Popular data analysis tools

	R	Python	Excel
Learning curve	Steeper learning curve for people without a programming background	Can be easy to learn for people without a programming background	Easy to learn for any analyst
Automation	Once you write commands you won't have to re-do the work, just upload a new data set	Once you write commands you won't have to re-do the work, just upload a new data set	New data sets are not always plug and play with your analysis
Analyzing data	Lots of libraries contributed by a broad user community	Over 6,500 packages contributed by the community including top academics	Limited to any particular version
Speed	In-memory, only limited by your computer's RAM	In-memory, only limited by your computer's RAM	Can be slower unless an enterprise configuration is used
Type of data	Reads data of almost any type	Reads data of almost any type	Limited to xlsx and csv files unless macros are used

	R	Python	Excel
Compatibility	Compatible with almost any data output, storage or processing platform	Not as compatible as Python with some data architecture systems, may need custom-built interfaces	While macros enhance compatibility, Excel is comparatively limited
Data manipulation	Very flexible data manipulation	Very flexible data manipulation, augmented by numerous data processing and manipulation packages	Color-coded formulas can be easier to use but have a more limited functionality
Seeing data	Command line presentation unless visualization libraries are used	Spreadsheet-like view function that can be less intuitive to navigate	Easy to navigate spreadsheet
Graphics	Cutting edge graphics, however advanced coding and JavaScript knowledge may be necessary	Cutting edge graphics including dynamic visualizations, maps, network graphs, etc.	More limited options based on pre-set drop-down menus (unless macros are used)
Cost & platform	Free, any platform	Free, any platform	Hundreds of dollars, functionality on a Mac does not always mimic a PC

Additional reading & reference

Doing Data Science by Cathy O'Neil & Rachel Schutt

Data Science for Business by Foster Provost & Tom Fawcett

Data Smart by John W. Foreman

Mining the Social Web by Matthew A. Russell

Predictive Analytics by Eric Siegel

Analyzing the Analyzers by Harlan Harris, Sean Murphy and Marck Vaisman

Use cases and additional examples

BNY Mellon advances artificial intelligence tech across operations

New York City uses "nudges" to reduce missed court dates

Proof of concept: Using predictive modeling to prioritize building inspections

Recruiting Chatbots in 2021: In-Depth Guide (updated 2026)

Tactical Institute: Protecting people and communities with pre-emptive social media threat analytics